

Problem 1.

C.

Problem 2.

a) D.

b) C.

Problem 3.

A.

Problem 4.

a) See solution document.

b) $v = \sqrt{2gR(1 + \cos \theta)}$

c) $\theta_c = 131.8^\circ$

Problem 5.

a) $d = 0.12 \text{ m}$

Problem 6.

D.

Problem 7.

a) $v = \sqrt{2gR(1 - \cos \theta)}$

b) $T = mg(3 - 2 \cos \theta)$

Problem 8.

a) $d = \frac{3}{5}L$